

Financial Stress Propagation in Semiconductor Supply Networks: A Network-Based Modeling Approach

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ABSTRACT

This project develops a network-based model of financial stress propagation in the semiconductor supply chain. Firm-level financial stress is measured using the Altman Z''-Score, while supply-chain relationships are represented as a directed weighted network of firms across key value-chain segments within the semiconductor industry. The model simulates how a financial shock at one firm or group of firms can spread through firm relationships, with propagation being determined by the relationship strength, firm centrality, and firm vulnerability/stress. Results are presented through an interactive dashboard that allows users to test and trigger shock scenarios, observing changes in firm stress. The analysis shows that financial risk in semiconductor supply chains cannot be assessed only at the individual level, as network position plays an important role in determining how stress spreads.

1. INTRODUCTION

Modern production is organised across globally distributed supply networks in which firms are tightly interdependent. Disruptions at one node propagate through the network, affecting firms with no direct exposure to the original shock. Analysing these propagation dynamics is therefore essential to understanding systemic risk at the industry level.

The semiconductor industry exemplifies this fragility. Advanced chip production relies on a narrow set of highly specialised suppliers—equipment from the Netherlands, photoresists from Japan, silicon wafers from Taiwan—with limited substitutability at each stage. Recent disruptions, from COVID-19 shortages to export control escalations, have demonstrated how this concentration generates systemic risk, with effects cascading into automotive, defence, and consumer electronics. As the foundational input of the digital economy, semiconductors represent an important case for supply chain risk analysis.

Financial stress—the impairment of a firm’s ability to meet its obligations—constitutes one such propagation channel. A distressed node may delay output, defer investment, or exit the market, imposing downstream costs on dependent firms and potentially triggering secondary stress events. This project investigates precisely this mechanism: how does financial stress propagate through the semiconductor supply chain, and which nodes are most systemically exposed?

The analysis identifies chokepoints and systemic vulnerabilities that firm-level assessment alone cannot reveal, under shock scenarios spanning economic, geopolitical, and environmental origins. Results are presented through an interactive dashboard enabling users to trigger shocks and observe stress propagation across the network in real time.

2. DATA COLLECTION & ASSUMPTIONS

Firm Selection

Our primary objective in selecting firms was to represent the semiconductor value chain as comprehensively as possible. The foundational reference for this mapping was the OECD report Mapping the Semiconductor Value Chain (June 2025), which provides a systematic overview of all stages of semiconductor production, including both core stages—design, fabrication, and packaging—and the supporting sectors that supply materials, chemicals, and equipment. From this report, we extracted the most significant companies mentioned at each stage.

For segments in which the OECD report did not identify specific firms, we supplemented the list using sector-specific market research reports covering EUV photoresists, high-purity wet chemicals, specialty gases, CMP slurry, and sputtering targets. The overall structure of the value chain was cross-validated against the CSIS report Mapping the Semiconductor Supply Chain: The Critical Role of the Indo-Pacific Region (May 2023), to ensure consistency in the identification of key stages and dependencies.

The process yielded a final list of 70 firms, all of which have publicly available financial filings. Each firm was assigned to its respective position within the value chain, enabling an assessment of stage-level dependencies and informing the subsequent network analysis.

Financial Data Collection

Firm-level financial stress is quantified using the Altman Z'' -Score (Altman, 1995), defined as:

$$Z'' = 6.56 \cdot X_1 + 3.26 \cdot X_2 + 6.72 \cdot X_3 + 1.05 \cdot X_4$$

where X_1 = Working Capital / Total Assets, X_2 = Retained Earnings / Total Assets, X_3 = EBIT / Total Assets, and X_4 = Book Equity / Total Liabilities.

The Z'' -Score was selected over alternative credit risk measures, including the original Altman Z-Score and the Ohlson O-Score, for two reasons. First, all four input variables are directly available from a company's financial statements, without requiring market capitalisation data or variables that are inconsistently reported across jurisdictions. Second, the model is applicable across industries and accounting standards, making it well-suited to a geographically diverse firm sample spanning the United States, Japan, Europe, Taiwan, and South Korea.

Given the international composition of the firm list, data collection was organised into three pipelines according to the filing regime of each company. For US-listed companies filing annual 10-K reports, financial data was retrieved programmatically via the SEC EDGAR XBRL API, which provides structured, machine-readable access to standardised financial concepts for all domestic public filers. For Japanese companies, data was collected through the EDINET DB API, which normalises financial data from Japan's EDINET disclosure system across both J-GAAP and IFRS filers. For all remaining companies—including firms filing 20-F reports with the SEC and companies listed on European, Taiwanese, and South Korean exchanges—financial data was extracted from the most recent publicly available annual reports using automated code-based extraction.

To enable comparison across firms, Z'' -scores were mapped to a normalised stress index in $[0, 1]$ using a logistic transformation. This approach was preferred over a linear

mapping as it assigns similar stress scores to firms with similar Z''-scores, while preserving meaningful differentiation at the extremes.

Relationship Definition

Inter-firm relationships were sourced from various sources, including annual reports, documented firm relationships, and further external sources. Each pair of firms connected within this dataset was treated as an edge in the supply chain network, with the edge weight reflecting the strength of the relationship.

Edge weights were determined by considering four factors: the relative prominence of the relationship; the market position of each firm within its segment; the nature of the relationship between the two firms; and the degree of substitutability of the supplier in the event of its sudden exit from the market. Relationships involving firms with dominant market positions and limited substitutes were assigned greater weight, reflecting their higher potential impact on the network in a stress scenario.

Two types of relationships are distinguished: buyer–supplier relationships, in which stress propagates unidirectionally from supplier to buyer, and partner relationships, which are modelled as bidirectional with half the propagation strength in each direction relative to a comparable buyer–supplier link.

Assumptions

Regarding financial data: the Z''-Score was selected in part because it eliminates the need for assumptions about missing or unavailable data. All four input variables are reported in the financial statements of any public company, regardless of the accounting standard applied. As all 70 firms in the sample have active public filings, no imputation or estimation of financial inputs was required.

Regarding supply chain relationships: the inter-firm links are assumed to accurately represent the structure of the semiconductor supply chain at the time of the analysis. Edge weights, as described earlier, are treated as fixed for the purposes of the baseline model.

3. DETERMINISTIC MODEL

The risk propagation model represents the semiconductor supply chain as a directed weighted network. Each firm is treated as a node, and each supplier-customer or partner relationship is represented as an edge. The purpose of the model is to simulate how financial stress at one firm can spread to other connected firms through supply-chain dependencies.

Stress Levels

Each firm is assigned a stress level, $s_i \in [0,1]$, where $s_i = 0$ represents no financial stress and $s_i = 1$ represents maximum financial distress. The initial stress levels are based on firm-level financial risk indicators, such as the adjusted Altman Z" score.

A financial shock is introduced by increasing the stress level of one firm, or a group of firms, depending on the chosen scenario. In practice, a scenario begins by selecting one or more initially shocked firms and assigning them an initial stress increase, which becomes Δs_i^0 in the first propagation round.

Propagation

Stress then propagates through the network over discrete rounds. In each round, a firm transmits only the new stress it received in the previous round. This prevents the same shock from being repeatedly counted without limit.

The propagation rule is as follows

$$\Delta s_j^{t+1} = \sum_i \alpha \cdot w_{ij} \cdot (1 + c_i) \cdot (1 + v_j) \cdot \Delta s_i^t,$$

and the stress level of the receiving firm is updated as

$$s_j^{t+1} = \min(1, s_j^t + \Delta s_j^{t+1}).$$

In this formula, Δs_j^{t+1} is the additional stress received by firm j in the next round. The parameter α is a global damping factor that controls how strongly shocks spread through the network. The term w_{ij} is the normalised dependency strength between firm i and firm j , based on the relationship score from 1 to 5. A stronger relationship means that stress from the source firm has a larger effect on the receiving firm.

The term c_i represents the centrality of the source firm. This captures the idea that shocks coming from highly connected or strategically important firms should have a stronger system-wide effect. The term v_j represents the vulnerability of the receiving firm, meaning that firms with weaker financial positions or more concentrated supplier bases are more sensitive to shocks.

The expressions $(1 + c_i)$ and $(1 + v_j)$ act as amplifiers. The reason for adding 1 is that centrality and vulnerability should increase the magnitude of propagation relative to a baseline, but they should not eliminate transmission completely when the value is close to zero. In other words, even a low-centrality firm can still transmit some stress, and even a less vulnerable firm can still be affected.

The model stops propagating stress when the maximum additional stress in a round becomes smaller than a chosen threshold, or when a fixed maximum number of rounds is reached. This makes the model deterministic: for the same input data, parameters, and shock scenario, it always produces the same output.

Shock Scenarios

In the current implementation, we consider three main shock scenarios. The first is an ASML idiosyncratic shock, where stress originates from ASML as a critical lithography bottleneck. The second is a silicon wafer sector shock, where stress is introduced to major wafer suppliers, reflecting a disruption in upstream wafer production. The third is a global credit tightening shock, where stress is applied more broadly across the network to represent a macro-financial deterioration affecting firms' access to credit. These scenarios allow us to compare how firm-specific, sector-specific, and system-wide shocks propagate through the semiconductor supply chain.

4. DASHBOARD

The dashboard translates the model outputs into an interactive network view, allowing users to observe how financial stress spreads through firm relationships. It supports the project's objective of simulating stress propagation across supply chains and analysing these dynamics to identify systemic risk.

Network Visualisation

In the network graph, each node represents a firm, while each edge represents a relationship between firms. Edge thickness reflects the strength or importance of the relationship: thicker links indicate stronger dependencies. Hovering over a node displays firm-specific information, including the firm name, value-chain segment, and stress levels.

The dashboard includes three main view modes. The Baseline view shows the initial, unshocked state of the network and serves as the reference point for comparison. In this view, firm stress levels are derived from the Altman Z''-Score. The Impact view shows the change in stress caused by the simulation, highlighting the firms most affected by the shock rather than those with the highest absolute stress. The Final view shows firm stress levels after the propagation process has completed. Together, these views distinguish between the initial condition, the marginal effect of the shock, and the final risk outcome.

Node color represents financial stress. In the Baseline and Final views, a green-to-red gradient is used, with green representing the least stressed firms and red representing the most stressed firms. In the Impact view, stress changes are shown using a gray-to-red gradient, where gray represents little or no change and red represents a large increase in stress.

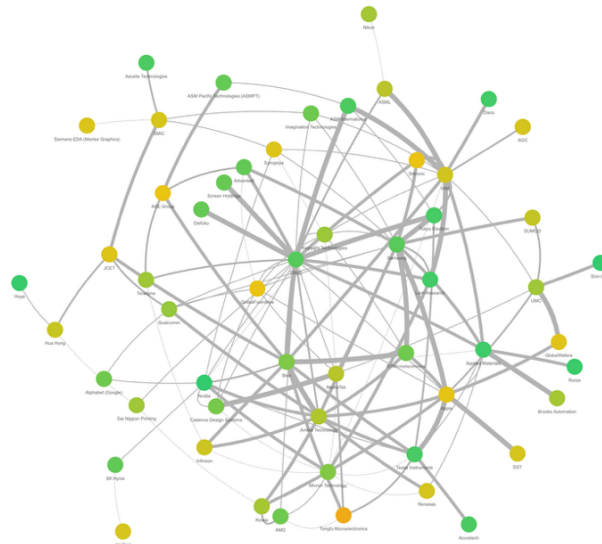


Figure 1: Baseline Stress

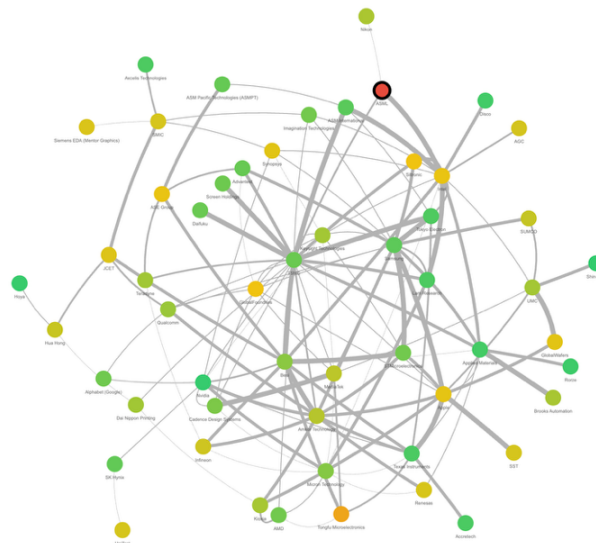


Figure 2: Final Stress

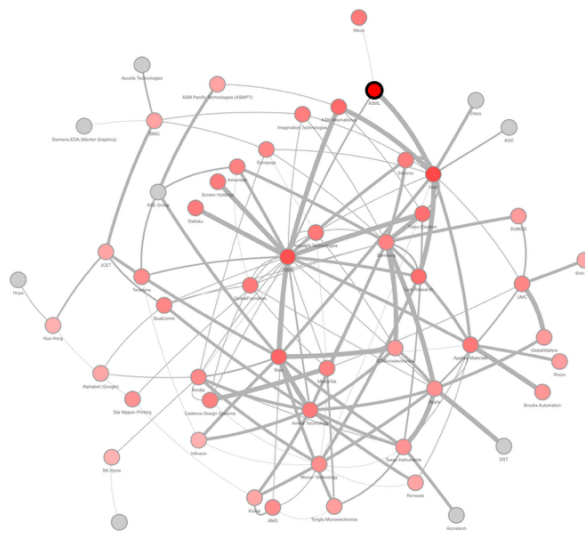


Figure 3: Impact (change in stress)

The three figures show an example scenario of an Idiosyncratic Shock on ASML (magnitude 0.5, alpha 0.15). Figure 1 shows the initial stress of the network, Figure 2 shows final stress, and Figure 3 shows the change in stress. The dashboard version allows users to zoom and inspect node-level labels interactively; the static figures are included only as illustrative snapshots.

Scenario Controls and Outputs

The control panel allows users to select the shock type, target firm or sector, shock magnitude, and propagation parameter alpha. Magnitude determines the severity of the initial shock, while alpha controls how strongly stress is transmitted through the network. Running a simulation updates both the network and the associated analytical views.

The analytical results tab summarises each scenario using indicators such as the number of propagation rounds, average final stress, average stress increase, and number of distressed firms. Additional charts rank firms by largest stress increase and systemic importance.

5. KEY INSIGHTS

The key insights derived from the model can be best understood by considering a few representative cases. Let us examine, for instance, a moderate idiosyncratic phenomenon (due to export restrictions to China) from ASML, a crucial supplier of many firms in the semiconductor industry.

Running a simulation of this shock (magnitude = 0.5 & alpha = 0.15), we can understand the true influence ASML has over firms in the semiconductor industry. Almost all nodes have been impacted—most, quite severely. A few notable mentions: Intel and TSMC both experienced changes in stress delta several multiples higher than other firms. This matches reality very closely: Indeed, TSMC and Intel are ASML's top 2 customers and are, to a significant extent (ASML is the sole provider of these machines), reliant on its Extreme Ultraviolet lithography machines that are required for advanced chip manufacturing.

Suppose now a sector-specific shock—exploding AI demand producing DRAM and NAND shortages. We model this crisis as one with a magnitude of 0.85 and alpha 0.5—highly distributive and quickly manifesting. The infographic turns bright red, suggesting that the shock had a severe impact on firms. Very few firms are impervious, for example, Disco and AGC. These firms are left relatively unimpacted due to their business models; both firms are critical players in AI-driven expansion, but instead of being the consumers of

DRAM and NAND technologies, they are producers of technology that permit consumers to utilise DRAM and NAND technologies.

Finally, let us reflect on a systematic shock originating from geopolitical uncertainty driven by U.S.-imposed export restrictions. The dashboard shows a broad increase in stress across the network, especially Applied Materials. This follows logically from the fact that Applied Materials sits right at the intersection of U.S. technology and Chinese manufacturing.

6. LIMITATIONS

Data Limitations

The analysis relies on publicly available financial filings and disclosed or inferred supply-chain relationships. Although all firms in the sample have available financial data, the differences in reporting standards across countries may reduce comparability. In addition, firms are not required to fully disclose customer-supplier relationships, only the most important ones. This means that some network relationships and weights rely on informed assumptions rather than exact data.

The network is also static. In reality, semiconductor supply chains change over time, which can be due to new contracts, restrictions due to geopolitics, or supplier substitution. The analysis is therefore limited to a point in time.

Financial Stress Measure

Financial stress is based on the Altman Z''-Score, which is useful because it can be calculated from accounting data across all firms included. However, it is a simplified proxy for credit risk. It cannot capture market expectations, liquidity access, refinancing risk, or real-time changes in a firm's conditions.

Model Limitations

The propagation process is deterministic and does not include uncertainty or randomness. The model also assumes that stress always spreads proportionally through supply-chain links, whereas real effects may depend on contracts, inventories, substitutability, timing, and the duration of the disruption.

Dependency strengths are also reduced to a 1-5 scoring system, which simplifies complex relationships. Finally, key parameters such as alpha, the centrality amplifier, and the vulnerability amplifier are not empirically calibrated, so different parameter choices may produce stronger or weaker propagation effects.



Interpretation

Due to these model limitations, the dashboard should be interpreted as an exploratory decision support tool rather than a predictive credit-risk model. Its value lies in comparing scenarios and highlighting where financial stress may concentrate across the semiconductor value chain.



DATA SOURCES

Center for Strategic and International Studies (2023): [*Mapping the semiconductor supply chain: The critical role of the Indo-Pacific region*](#)

OECD (2025): [*Mapping the semiconductor value chain*](#)

Intel Market Research (2025): [*EUV photoresists market*](#)

Intel Market Research (2025): [*High purity semiconductor wet chemicals market*](#)

Intel Market Research (2025): [*Semiconductor specialty gases market 2025–2032*](#)

Research and Markets (2025): [*Chemical mechanical planarization \(CMP\) slurry market*](#)

Market Reports World (2025): [*Semiconductor sputtering targets market*](#)

DASHBOARD & CODE LINKS

[*Dashboard*](#)

[*GitHub Repository*](#)